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RESOLUTION MSC.137(76)
(adopted on 4 December 2002)
STANDARDS FOR SHIP MANOEUVRABILITYMSC 76/23/Add.1
ANNEX 6
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ANNEX

STANDARDS FOR SHIP MANOEUVRABILITY

1 P RINCIPLES
1.1 The Standards for ship manoeuvrability (the St andards) should be used to evaluate the manoeuvring performance of ships and to assist those responsible for the design, construction, repair and operation of ships.

1.2 It should be noted that the Standards were developed for ships with traditional propulsion and steering systems (e.g. shaft driven ships w ith conventional rudders). Therefore, the Standards and methods for establishing compliance may be periodically reviewed and updated by the Organization, as appropriate, taking in to account new technologies, research and development, and the results of experience with the present Standards.

2 G ENERAL

2.1 The Standards contained in this document are based on the understanding that the manoeuvrability of ships can be evaluated from the characteristics of conventional trial manoeuvres.

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2 G ENERAL

2.1 The Standards contained in this document are based on the understanding that the manoeuvrability of ships can be evaluated from the characteristics of conventional trial manoeuvres. The following two methods can be used to demonstrate compliance with these Standards:

.1 scale model tests and/or computer pred ictions using mathematical models can be performed to predict compliance at the design stage. In this case full-scale trials should be conducted to validate these results . The ship should then be considered to meet these Standards regardless of full-scale trial results, except where the Administration determines that the prediction efforts were substandard and/or the ship performance is in substantial disagreement with these Standards; and

.2 the compliance with the Standards can be demonstrated based on the results of the full-scale trials conducted in accordance with the Standards. If a ship is found in substantial disagreement with the Standard s, then the Administration should take remedial action, as appropriate.

3 A PPLICATION

3.1 Notwithstanding the points raised in paragr aph 1.2 above, the Standards should be applied

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substantial disagreement with the Standard s, then the Administration should take remedial action, as appropriate.

3 A PPLICATION

3.1 Notwithstanding the points raised in paragr aph 1.2 above, the Standards should be applied to ships of all rudder and propulsion types, of 100 m in length and over, and chemical tankers and gas carriers regardless of the length.

3.2 In the event that the ships referred to in paragraph 3.1 above undergo repairs, alterations or modifications, which, in the opinion of the Administration, may influence their manoeuvrability characteristics, the continued compliance with the Standards should be verified.

3.3 Whenever other ships, originally not subject to the Standards, undergo repairs, alterations or modifications, which, in the opinion of the Admini stration, are of such an extent that the ship may be considered to be a new ship, then th at ship should comply with these Standards. Otherwise, if the repairs, alterations and modifications, in the opinion of the Administration, may influence the manoeuvrability characteristics, it should be demonstrated that these characteristics

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Otherwise, if the repairs, alterations and modifications, in the opinion of the Administration, may influence the manoeuvrability characteristics, it should be demonstrated that these characteristics do not lead to any deterioration of the manoeuvrability of the ship.

3.4 The Standards should not be applied to high-speed craft as defined in the relevant Code.

4 D EFINITIONS

4.1 Geometry of the ship

4.1.1 Length (L) is the length measured between the aft and forward perpendiculars.

4.1.2 Midship point is the point on the centreline of a ship midway between the aft and forward perpendiculars.

4.1.3 Draught (Ta) is the draught at the aft perpendicular.

4.1.4 Draught (Tf) is the draught at the forward perpendicular.

4.1.5 Mean draught (Tm) is defined as Tm = (Ta + Tf)/2.

4.1.6 Trim () is defined as = (Ta - Tf).

4.1.7 is the full load displacement of the ship (tonnes).

4.2 Standard manoeuvres and associated terminology

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4.1.7 is the full load displacement of the ship (tonnes).

4.2 Standard manoeuvres and associated terminology

Standard manoeuvres and associated terminology are as defined below:

.1 The test speed (V) used in the Standards is a speed of at least 90% of the ships speed corresponding to 85% of the maximum engine output.

.2 Turning circle manoeuvre is the manoeuvre to be performed to both starboard and port with 35° rudder angle or the maximu m rudder angle permissible at the test speed, following a steady approach with zero yaw rate.

.3 Advance is the distance travelled in the direction of the original course by the midship point of a ship from the position at which the rudder order is given to the position at which the heading has changed 90° from the original course.

.4 Tactical diameter is the distance travelle d by the midship point of a ship from the position at which the rudder order is given to the position at which the heading has

changed 180° from the original course. It is measured in a direction perpendicular

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position at which the rudder order is given to the position at which the heading has changed 180° from the original course. It is measured in a direction perpendicular to the original heading of the ship.

.5 Zig-zag test is the manoeuvre wher e a known amount of helm is applied alternately to either side when a know n heading deviation from the original heading is reached.

.6 The 10°/10° zig-zag test is performed by turning the rudder alternately by 10° to either side following a heading deviation of 10° from the original heading in accordance with the following procedure: